

ZIRUI YAN

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SUMMARY

Duolingo AI Research Engineer II

Ph.D. in Electrical, Computer, and Systems Engineering (ECSE); Dual B.S. in Statistics and B.E. in CS.

Expertise: causal bandits, machine learning, large language models.

INDUSTRY EXPERIENCE

Duolingo

New York, NY

AI Research Engineer II, Monetization Eigen Team

Jun 2026 - Present

Working on extending bandit to mid-session no energy case to boost the subscription

Duolingo

Pittsburgh, PA

AI Research Engineer Intern, Monetization Eigen Team (Mentor: Badr Albanna)

May 2025 - Aug 2025

Accelerated training and deployment pipeline by >40% on **BigQuery and Vertex AI** in Google Cloud

Pre-processed billions of ad history records using **Polars** for v2 experiment in <1h

Added **temporal features** that capture ad interactions and event-time dependence

Launched **new ads bandit experiment**, boosting session-end conversion rate by 10% (\$35k/day in revenue)

Yahoo! Research

Remote

Research Intern, Search and Recommendation (Mentor: Xinyue Wang, Manager: Rao Shen)

May 2024 - Aug 2024

Worked on Yahoo! search query understanding for downstream tasks

Generated pseudo-labels with >90% accuracy via **zero-shot prompting**

Achieved >70% knowledge-transfer accuracy to **compact BERT models** through data augmentation

IBM Research

Yorktown Heights, NY

Research Extern, Trustworthy AI (Mentor: Tian Gao)

May 2023 - Aug 2023

Developed efficient **bandit-based prompt selection** algorithms for large language models (LLM)

Addressed non-stationary, stochastic reward settings, and outperformed RL baselines by 10% in LLMs

EDUCATION

Rensselaer Polytechnic Institute (RPI)

Troy, NY

Ph.D., Electrical, Computer, and Systems Engineering

2026

Dissertation: Causal Bandits with Soft Intervention: A Unified Framework (Advisor: Prof. Ali Tajer)

University of Science and Technology of China (USTC)

Hefei, China

B.S. Statistics & B.E. Computer Science and Technology

2016 - 2020

PUBLICATIONS (* equal contribution)

Peer reviewed papers

Causal Bayesian Optimization via Causal Bandits: Open Questions

A. Mukherjee, **Z. Yan**, A. Tajer

Proc. UAI Workshop on Causality for Decision Making (**UAI CDM**) 2026.

Multi-component Causal Tracing in Large Language Models

Z. Yan, D. Wei, D. A. Katz, P. Sattigeri, A. Tajer

Proc. Association for Computational Linguistics (**ACL**) 2026. Oral presentation.

Reward-oriented Causal Representation Learning

Z. Yan*, E. Acartürk*, A. Tajer

Proc. Conference on Neural Information Processing Systems (**NeurIPS**) 2025

Linear Causal Bandits: Unknown Graph and Soft Interventions

Z. Yan, A. Tajer

Proc. Conference on Neural Information Processing Systems (**NeurIPS**) 2024

Improved Bound for Robust Causal Bandits with Linear Models

Z. Yan, A. Mukherjee, B. Varıcı, A. Tajer

Proc. International Symposium on Information Theory (**ISIT**) 2024

Causal Bandits with General Causal Models and Interventions

Z. Yan, D. Wei, D. A. Katz, P. Sattigeri, A. Tajer

Proc. International Conference on Artificial Intelligence and Statistics (**AISTATS**) 2024

Robust Causal Bandits for Linear Models

Z. Yan, A. Mukherjee, B. Varıcı, A. Tajer

IEEE Journal on Selected Areas in Information Theory (**JSait**) 2024

Federated multi-armed bandit via uncoordinated exploration

Z. Yan, Q. Xiao, T. Chen, A. Tajer

Proc. International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2022

Optimizing Parameter Mixing Under Constrained Communications in Parallel Federated Learning

X. Liu*, **Z. Yan***, Y. Zhou, D. Wu, X. Chen, J. H. Wang

IEEE/ACM Transactions on Networking (**TON**), 2023

Image denoising via K-SVD with primal-dual active set algorithm

Q. Xiao, C. Wen, **Z. Yan**

Proc. Winter Conference on Applications of Computer Vision (**WACV**), 2020

PROFESSIONAL SERVICE

Conference reviewer: ICML (2025), ICLR (2025), NeurIPS (2024–2026), AISTATS (2024), and ISIT; NeurIPS 2024 Top Reviewer.

Journal reviewer: IEEE Transactions on Signal Processing, Transactions on Machine Learning Research (2025), and IEEE Journal on Selected Areas in Information Theory.

Workshop reviewer: ICML 2023 Workshop on Structured Probabilistic Inference and Generative Modeling (SPIGM).

SKILLS

Programming Languages: Python, SQL, R, LaTeX, Markdown

Software: PyTorch, vLLM, Hugging Face, pandas/Polars, DBT, Vertex AI, BigQuery

Systems: Docker, Kubernetes